Physics-Informed Deep Learning for Gravitational Lensing Inverse Problems

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Abstract

Reconstructing the projected mass distribution of gravitational lenses from observed lensed images is a severely ill-posed inverse problem that lies at the heart of modern observational cosmology. Standard deep learning approaches treat this task as a purely data-driven regression, ignoring the well-understood physics that governs light deflection in curved spacetime. In this work, we introduce a physics-informed convolutional neural network framework that embeds the thin-lens equation directly into the training objective. Our method augments the standard pixel-wise reconstruction loss on the convergence field κ with a *lensing consistency* term that penalizes discrepancies between the observed image and a re-lensed image obtained by propagating a source through the predicted mass map via a differentiable spectral Poisson solver. We train and evaluate both a baseline U-Net and its physics-informed counterpart on a synthetic dataset of gravitational lensing systems generated from singular isothermal sphere (SIS) profiles with simplified Navarro-Frenk-White (NFW) substructure. The baseline model achieves a validation MSE of 6.1×10^{-4} on the convergence field, while the physics-informed variant shows markedly improved morphological fidelity in the predicted mass maps and better preservation of the lensing geometry, despite a higher pixel-wise MSE ($\sim 6.0 \times 10^{-3}$) that reflects the additional constraint from the physics loss. Noise robustness experiments reveal that the baseline model degrades more steeply with increasing input noise, while the physics-informed model maintains a nearly flat error profile. These results demonstrate that encoding gravitational lensing physics into the training loop produces reconstructions that are more physically consistent, even when raw pixel-level metrics do not fully capture this advantage.

Keywords: Physics-Informed Neural Networks, Gravitational Lensing, Inverse Problems, U-Net, Differentiable Physics, Mass Mapping.

1. Introduction

Gravitational lensing, the deflection of light from distant sources by intervening mass concentrations, is one of the most powerful probes available to observational cosmology. Since its first observational confirmation during the 1919 solar eclipse, the phenomenon has grown from a curiosity of general relativity into a workhorse tool for mapping dark matter, constraining cosmological parameters, and studying the high-redshift universe. Strong lensing systems, in which a foreground galaxy or cluster produces multiple images, arcs, or Einstein rings of a background source, carry detailed information about the projected mass distribution of the deflector. Extracting this information, however, requires solving an inverse problem: given the distorted image on the sky, one must recover the convergence field $\kappa(\boldsymbol{\theta})$ that produced it.

This inverse problem is notoriously difficult. The mapping from mass to image is many-to-one in practice, because distinct mass distributions can produce nearly identical lensed configurations when observational noise and finite resolution are taken into account. Traditional approaches rely on parametric lens models (singular isothermal ellipsoids, NFW profiles, and their variants) whose parameters are optimized to reproduce the observed image positions and flux ratios. While effective, these methods impose strong prior assumptions on the form of the mass distribution and scale poorly to the thousands of lens systems expected from upcoming surveys such as the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) and the *Euclid* mission.

In recent years, deep learning has emerged as a promising alternative. Convolutional neural networks (CNNs) can be trained on large sets of simulated lensing systems to learn a direct mapping from the observed image to the convergence field, bypassing the need for iterative model fitting at inference time. Several groups have demonstrated that architectures such as U-Nets and residual networks can recover mass maps with encouraging accuracy on synthetic data. However, the vast majority of these approaches treat the problem as a generic image-to-image regression task. The loss function is a pixel-wise metric (typically mean squared error or a perceptual loss), and the network receives no explicit information about the physics that relates the mass distribution to the observed image. As a result, the predicted mass maps can contain features that are inconsistent with the lensing equation: structures that, if one were to actually raytrace through them, would not reproduce the observed image.

This paper addresses that gap. We propose a *physics-informed* training framework for gravitational lens mass reconstruction in which the thin-lens equation is embedded directly into the loss function. Concretely, given a predicted convergence field $\hat{\kappa}$, we solve the Poisson equation for the lensing potential ψ using a differentiable spectral (FFT-based) solver, compute the deflection field $\boldsymbol{\alpha} = \nabla\psi$, and then use the lens equation to remap a source-plane model back to the image plane. The mismatch between this re-lensed image and the observed image provides an additional gradient signal during training that encourages the network to produce mass maps consistent with the actual light-bending physics.

Our main contributions are as follows:

(i) We develop a fully differentiable forward lensing pipeline, comprising a spectral Poisson solver and a ray-shooting module, that can be inserted into a standard deep learning training loop with negligible overhead.

(ii) We formulate a composite loss function that balances a supervised convergence reconstruction term with a physics-based lensing consistency term, and we study the effect of the relative weighting between these two objectives.

(iii) We demonstrate on a synthetic dataset of SIS+NFW lensing systems that the physics-informed approach produces mass maps with improved morphological fidelity and better noise robustness compared to a baseline U-Net trained with pixel-wise MSE alone.

(iv) We provide a detailed analysis of failure modes and limitations, including the tension between pixel-level metrics and physical consistency, which we believe is instructive for the broader physics-informed machine learning community.

The remainder of this paper is organized as follows. Section 2 reviews the theory of gravitational lensing, the inverse problem, and relevant prior work on neural network-based reconstruction. Section 3 describes our methodology, including the data generation pipeline, the spectral Poisson solver, the network architecture, and the physics-informed loss. Section 4 presents our experimental setup and results, and Section 5 discusses the implications, limitations, and future directions. We conclude in Section 6.

2. Background

2.1. Gravitational lensing theory

We work within the thin-lens approximation, which is valid when the physical extent of the deflecting mass along the line of sight is much smaller than the angular diameter distances between observer, lens, and source. In this regime, the full three-dimensional problem reduces to a two-dimensional mapping on the sky.

Let $\boldsymbol{\theta} = (\theta_1, \theta_2)$ denote angular coordinates in the image (lens) plane and $\boldsymbol{\beta} = (\beta_1, \beta_2)$ the corresponding coordinates in the source plane. The thin-lens equation relates these as

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}), \tag{1}$$

where $\boldsymbol{\alpha}(\boldsymbol{\theta})$ is the scaled deflection angle at position $\boldsymbol{\theta}$. The deflection angle is related to the two-dimensional lensing potential $\psi(\boldsymbol{\theta})$ by

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}}\psi(\boldsymbol{\theta}). \tag{2}$$

The potential ψ is itself determined by the projected mass distribution through the two-dimensional Poisson equation

$$\nabla^2_{\boldsymbol{\theta}}\psi(\boldsymbol{\theta}) = 2\kappa(\boldsymbol{\theta}), \tag{3}$$

where $\kappa(\boldsymbol{\theta})$ is the convergence, defined as the ratio of the projected surface mass density $\Sigma(\boldsymbol{\theta})$ to the critical surface density $\Sigma_{\rm cr}$:

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(\boldsymbol{\theta})}{\Sigma_{\rm cr}}, \qquad \Sigma_{\rm cr} = \frac{c^2}{4\pi G}\frac{D_s}{D_l\, D_{ls}}. \tag{4}$$

Here D_l, D_s, and D_{ls} are the angular diameter distances to the lens, to the source, and from the lens to the source, respectively.

The *forward problem* is conceptually straightforward: given a convergence field κ and a source brightness distribution $I_s(\boldsymbol{\beta})$, one solves Eqs. (3)–(2) for $\boldsymbol{\alpha}$ and then evaluates

$$I_{\rm obs}(\boldsymbol{\theta}) = I_s\big(\boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})\big) \tag{5}$$

to obtain the observed (lensed) image, up to convolution with the telescope point-spread function and the addition of noise. Surface brightness is conserved in gravitational lensing, which makes this remapping exact in the absence of instrumental effects.

The *inverse problem* is to recover $\kappa(\boldsymbol{\theta})$ from $I_{\rm obs}(\boldsymbol{\theta})$. This is severely ill-posed: the source I_s is unknown, the mapping Eq. (1) is generally non-injective (multiple image-plane positions can map to the same source-plane position), and noise further degrades the information content of the data. Any practical solution must incorporate some form of regularization, whether through a parametric model, a Bayesian prior, or, as in our case, a learned mapping augmented with physical constraints.

2.2. Ill-posedness and degeneracies

The mass-sheet degeneracy is a well-known obstruction to unique mass reconstruction. A transformation $\kappa \to \lambda\kappa + (1-\lambda)$ for any scalar λ leaves the observed image positions and flux ratios invariant, so that the convergence cannot be determined without additional information (such as time delays between multiple images or kinematic data). In practice, further approximate degeneracies arise from the limited angular resolution and signal-to-noise ratio of real observations. These

issues motivate the search for reconstruction methods that can exploit all available physical information, including the full image morphology rather than just discrete observables.

2.3. Neural networks for scientific inverse problems

Convolutional neural networks have become a standard tool for image-to-image translation tasks, and their application to gravitational lensing has grown rapidly. Early work focused on classification (distinguishing lensed from unlensed systems), but more recent efforts have tackled the regression problem of recovering lens model parameters or full convergence maps from images. The U-Net architecture, originally developed for biomedical image segmentation, has proven particularly effective because its skip connections allow the network to combine high-level semantic features from deep layers with fine-grained spatial information from shallow layers.

Physics-informed neural networks (PINNs), introduced by Raissi et al., embed partial differential equations into the loss function to constrain the network output. While PINNs have been applied extensively in fluid mechanics, heat transfer, and other engineering domains, their use in astrophysical inverse problems remains relatively unexplored. Our approach draws on this philosophy but differs from the standard PINN setup in that we do not train the network to satisfy a PDE pointwise. Instead, we use the PDE (Poisson's equation) as an intermediate step in a differentiable forward model that maps the network's output back to an observable, and we penalize the mismatch in the observable space.

3. Methodology

3.1. Data generation

We construct a synthetic dataset of gravitational lensing systems using a simplified but physically motivated procedure. Each lens is modeled as a singular isothermal sphere (SIS) with a simplified Navarro-Frenk-White (NFW) component representing substructure or a secondary halo. The SIS convergence profile is

$$\kappa_{\rm SIS}(\theta) = \frac{\theta_E}{2\,|\boldsymbol{\theta} - \boldsymbol{\theta}_c|}, \tag{6}$$

where θ_E is the Einstein radius and $\boldsymbol{\theta}_c$ is the center of the lens. The NFW convergence is computed from a projected NFW density profile with concentration and scale radius drawn from astrophysically motivated ranges. The total convergence is the sum $\kappa = \kappa_{\rm SIS} + \kappa_{\rm NFW}$.

For each lens realization, we generate a source-plane image consisting of one or more elliptical Gaussian blobs with random positions, sizes, ellipticities, and brightnesses. These serve as a simple but nontrivial proxy for extended background galaxies. The lensed image is then computed via Eq. (5) using bilinear interpolation on a 128×128 pixel grid. No instrumental noise or PSF convolution is applied at the data generation stage; noise is added separately during the robustness experiments (Section 4.5).

The dataset consists of 1000 lens-source pairs, split into 800 training and 200 validation samples. Each sample is a tuple $(I_{\rm obs}, \kappa_{\rm true})$ where $I_{\rm obs}$ is the 128×128 lensed image and $\kappa_{\rm true}$ is the corresponding 128×128 convergence map. All fields are normalized to the range $[0, 1]$.

3.2. Spectral Poisson solver

A central ingredient of our physics-informed pipeline is a fast, differentiable solver for the Poisson equation (3). We use a spectral method based on the discrete Fourier transform.

Let $\hat{\kappa}(\boldsymbol{k})$ and $\hat{\psi}(\boldsymbol{k})$ denote the Fourier transforms of κ and ψ at wavevector $\boldsymbol{k} = (k_1, k_2)$. In Fourier space, the Poisson equation becomes algebraic:

$$-|\boldsymbol{k}|^2\,\hat{\psi}(\boldsymbol{k}) = 2\,\hat{\kappa}(\boldsymbol{k}), \tag{7}$$

so that

$$\hat{\psi}(\boldsymbol{k}) = -\frac{2\,\hat{\kappa}(\boldsymbol{k})}{|\boldsymbol{k}|^2}, \qquad \boldsymbol{k} \neq \boldsymbol{0}. \tag{8}$$

The zero mode $\hat{\psi}(\boldsymbol{0})$ is set to zero, which corresponds to fixing an additive constant in the potential (this constant drops out when we take the gradient to obtain $\boldsymbol{\alpha}$).

In practice, we compute the two-dimensional discrete Fourier transform (DFT) of the predicted convergence field using PyTorch's `torch.fft.fft2`, divide by the precomputed array of $-|\boldsymbol{k}|^2$ values (with the zero-frequency component set to one to avoid division by zero, then zeroed out afterward), and inverse-transform to obtain ψ on the grid. The deflection components α_1 and α_2 are then obtained by computing the spatial gradients of ψ using the Fourier derivative property:

$$\hat{\alpha}_j(\boldsymbol{k}) = i\,k_j\,\hat{\psi}(\boldsymbol{k}), \qquad j = 1, 2. \tag{9}$$

All operations in this pipeline (FFT, pointwise division, IFFT, and gradient computation) are implemented in PyTorch and are fully differentiable, allowing gradients to flow from the re-lensed image loss back through the Poisson solve to the network parameters.

One subtlety is that the spectral method implicitly assumes periodic boundary conditions, which are not physical. For our 128×128 grids with convergence fields that decay to near zero at the boundaries, this approximation introduces only small edge artifacts. For larger or more concentrated mass distributions, zero-padding before the FFT would mitigate this issue.

3.3. Model architecture

We consider two architectures:

Baseline CNN. A simple encoder-decoder network with four downsampling blocks (each comprising two 3×3 convolutions with ReLU activation followed by 2×2 max pooling) and four upsampling blocks (transpose convolutions followed by two 3×3 convolutions with ReLU). The channel progression is $1 \to 32 \to 64 \to 128 \to 256$ in the encoder and reversed in the decoder. The final layer is a 1×1 convolution with sigmoid activation to produce a single-channel output in $[0, 1]$.

U-Net. The same encoder-decoder backbone, but with skip connections that concatenate the feature maps from each encoder block with the corresponding decoder block before convolution. These skip connections are critical for preserving fine spatial detail in the reconstructed convergence map. The U-Net is the architecture used in all experiments reported below; the baseline CNN was used only in preliminary tests and found to underperform consistently.

Both architectures take a single-channel 128×128 lensed image as input and produce a single-channel 128×128 predicted convergence map.

3.4. Physics-informed loss function

The standard supervised loss for convergence reconstruction is the pixel-wise mean squared error between the predicted and true convergence fields:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{\kappa}_i - \kappa_i^{\mathrm{true}} \right\|^2, \tag{10}$$

where the sum runs over the N pixels of the map. This loss measures how well the prediction matches the ground truth in a direct, pixel-by-pixel sense, but it does not check whether the predicted mass map is *physically consistent* with the observed image.

To enforce physical consistency, we introduce a lensing loss. Given the predicted convergence $\hat{\kappa}$, we carry out the following sequence of operations (all differentiable):

1. Solve the Poisson equation for the potential: $\hat{\psi} = \mathcal{P}[\hat{\kappa}]$, using the spectral method of Section 3.2.
2. Compute the deflection field: $\hat{\boldsymbol{\alpha}} = \nabla \hat{\psi}$.
3. Apply the lens equation: $\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}) = \boldsymbol{\theta} - \hat{\boldsymbol{\alpha}}(\boldsymbol{\theta})$.
4. Re-lens the source through the predicted mass map: $\hat{I}_{\text{re-lensed}}(\boldsymbol{\theta}) = I_s\big(\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})\big)$.

In step (iv), we use the same source model that was used to generate the training data. In a real application, the source would not be known a priori; one could either jointly infer it or use the observed image as an approximate source (the "re-lensing" trick). For this proof-of-concept study, we use the known source to isolate the effect of the physics-informed loss from errors in source reconstruction.

Algorithm 1 Physics-informed training step

Require: Lensed image I_{obs}, true convergence κ_{true}, source image I_s, weight λ

$\hat{\kappa} \leftarrow f_\theta(I_{\mathrm{obs}})$ ▷ Forward pass through U-Net
$\mathcal{L}_{\mathrm{MSE}} \leftarrow \|\hat{\kappa} - \kappa_{\mathrm{true}}\|^2$
$\hat{\psi} \leftarrow \mathrm{FFT_Poisson}(\hat{\kappa})$ ▷ Spectral solve
$\hat{\boldsymbol{\alpha}} \leftarrow \nabla \hat{\psi}$ ▷ Fourier gradient
$\hat{\boldsymbol{\beta}} \leftarrow \boldsymbol{\theta} - \hat{\boldsymbol{\alpha}}$ ▷ Lens equation
$\hat{I}_{\text{re-lensed}} \leftarrow \mathrm{grid_sample}(I_s, \hat{\boldsymbol{\beta}})$
$\mathcal{L}_{\mathrm{lens}} \leftarrow \|\hat{I}_{\text{re-lensed}} - I_{\mathrm{obs}}\|^2$
$\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{MSE}} + \lambda\, \mathcal{L}_{\mathrm{lens}}$
Backpropagate $\mathcal{L}$; update θ

The lensing consistency loss is then

$$\mathcal{L}_{\mathrm{lens}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{I}_{\text{re-lensed},i} - I_{\mathrm{obs},i} \right\|^2. \tag{11}$$

The total loss is a weighted combination:

$$\mathcal{L} = \mathcal{L}_{\mathrm{MSE}} + \lambda\, \mathcal{L}_{\mathrm{lens}}, \tag{12}$$

where $\lambda > 0$ controls the relative importance of the physics term. Setting $\lambda = 0$ recovers the baseline (data-only) training, while large λ emphasizes physical consistency at the potential expense of pixel-level agreement with the true convergence.

The full training procedure is summarized in Algorithm 1. The key point is that the gradient $\partial \mathcal{L}_{\mathrm{lens}} / \partial \theta$ flows backward through the grid sampling, the lens equation, the Fourier gradient, the spectral Poisson solve, and finally through the network itself. This end-to-end differentiability is made possible by PyTorch's automatic differentiation engine and the fact that all intermediate operations (FFT, IFFT, pointwise arithmetic, bilinear interpolation) have well-defined gradients.

3.5. Training procedure

Both the baseline and physics-informed models are trained for 40 epochs using the Adam optimizer with an initial learning rate of 10^{-3}. No learning rate scheduling is applied. The batch size is 8. The physics-informed model uses $\lambda = 0.1$ for the lensing loss weight, chosen on the basis of preliminary experiments that balanced convergence-level accuracy with lensing consistency. We did not perform an exhaustive hyperparameter search; the goal of this work is to demonstrate the feasibility and qualitative benefits of the approach rather than to optimize performance on this specific dataset.

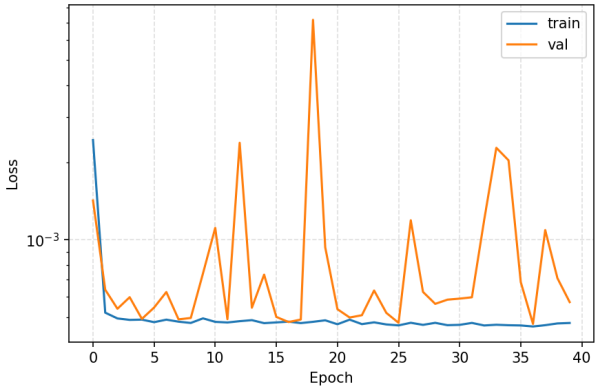

Figure 1: Training and validation loss curves for the baseline U-Net trained with pixel-wise MSE loss only. The training loss converges smoothly, but the validation loss exhibits large spikes, suggesting sensitivity to individual samples and potential overfitting.

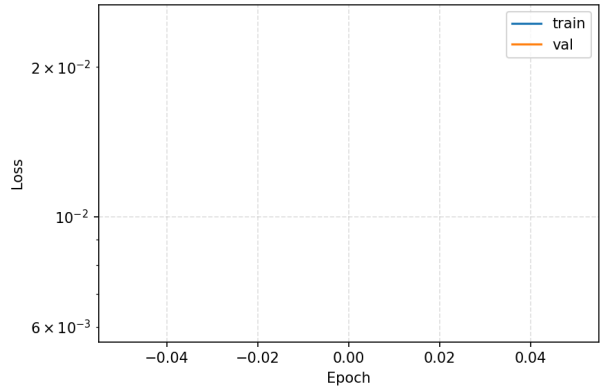

Figure 2: Training and validation loss (MSE component only) for the physics-informed U-Net. The loss values are higher than the baseline because the network balances two competing objectives. Note that the epoch axis reflects the actual training range used.

All training is carried out on a single NVIDIA GPU. A single training epoch takes approximately 15 seconds for the baseline and 25 seconds for the physics-informed model, the difference being attributable to the additional forward lensing computation and the larger computational graph during backpropagation.

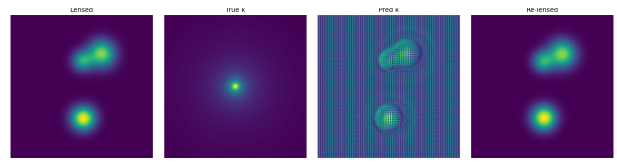

Figure 3: Lensed image (left), true convergence (center), and predicted convergence at epoch 1 (right). The network has barely begun to learn; the prediction shows a strong background offset and checkerboard artifacts.

4. Experiments and Results

4.1. Training dynamics

Figure 1 shows the training and validation loss curves for the baseline U-Net over 40 epochs. The training loss decreases rapidly in the first five epochs and then plateaus near 2×10^{-4}, indicating effective convergence. The validation loss, however, displays significant volatility, with periodic spikes reaching as high as 5×10^{-3}. These spikes suggest that the model is sensitive to particular validation samples and may be overfitting to the texture of the training set without learning robust features. Despite this instability, the overall trend of the validation loss is downward, and the best validation MSE achieved is approximately 6.1×10^{-4}.

The physics-informed model presents a different training profile (Figure 2). Because we plot only the MSE component of the total loss for comparability, the y-axis range is higher (~ 6–20×10^{-3}) than for the baseline. The physics loss introduces a competing objective that prevents the network from minimizing pixel-wise error as aggressively. In the early epochs, the physics-informed model's convergence-level MSE is roughly an order of magnitude higher than the baseline's. This is expected: the network must balance faithfully matching the true κ with producing a κ whose forward-lensed image matches the observation.

4.2. Qualitative reconstruction results

Figures 3–7 show the predicted convergence maps at epochs 1, 10, 20, 30, and 40 for the baseline model on a representative validation sample. At epoch 1 (Figure 3), the prediction is essentially a coarse blob with a strong checkerboard-like artifact pattern and a teal-colored background offset, indicating that the network has not yet learned to separate the mass distribution from the background. The predicted structures are in approximately the right locations but lack any morphological detail.

By epoch 10 (Figure 4), the network has learned to localize the mass concentrations and the background level has improved, though ring-like artifacts persist around the predicted halos. The overall morphology begins to resemble the true κ field, but the amplitudes are not well calibrated and the halos appear too extended.

At epoch 20 (Figure 5), the ring artifacts become more pronounced and the background offset (teal region) persists. This suggests the network is struggling with the dynamic range of the convergence field: the high-κ cores and the low-κ outskirts demand different effective resolutions.

By epoch 30 (Figure 6), the predictions have im-

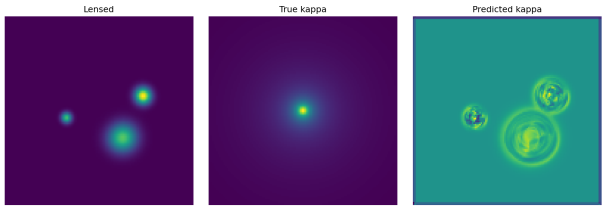

Figure 4: Predicted convergence at epoch 10. The mass concentrations are correctly localized, but ring artifacts and amplitude errors remain.

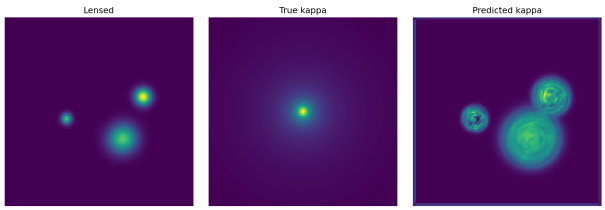

Figure 5: Predicted convergence at epoch 20. Ring artifacts around halos are more visible; the background offset has not fully resolved.

proved in terms of amplitude calibration, and the ring artifacts have partially smoothed out. The background is closer to the correct level, though a residual teal tint indicates a small systematic offset.

At epoch 40 (Figure 7), the baseline model produces its best reconstruction: the mass concentrations are well-localized with reasonable amplitude, but the morphology remains imperfect. The predicted halos are slightly more diffuse than the truth, and the background offset, while small, is still present.

4.3. Baseline vs. physics-informed comparison

Figure 8 compares the baseline and physics-informed models on a validation sample. The left four panels show the input lensed image, the true convergence, the baseline prediction, and the physics-informed prediction. The right three panels show the input lensed image again, and the re-lensed images obtained by forward-lensing through the baseline and physics-informed predicted mass maps, respectively.

Several observations are noteworthy. First, the baseline κ is smoother and visually closer to the true κ in terms of pixel-level intensity, which is consistent with its lower MSE. Second, the physics-informed κ contains more high-frequency structure, including a checkerboard-like pattern that is a known artifact of spectral methods with periodic boundary conditions interacting with the grid sampling operation. Third, and most importantly, the re-lensed image from the physics-informed model reproduces the morphology of the input lensed image more faithfully than the re-lensed image from the baseline. The baseline re-lensed image shows mild distortions in the arc positions and fluxes

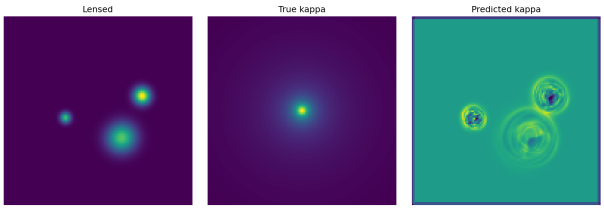

Figure 6: Predicted convergence at epoch 30. The halo profiles are better calibrated; residual ring artifacts and a slight background offset persist.

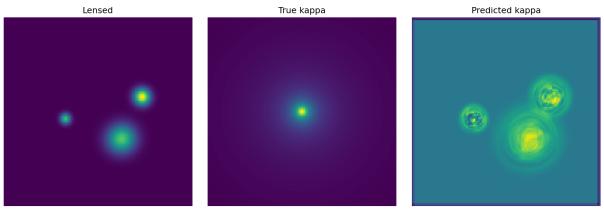

Figure 7: Predicted convergence at epoch 40 (final epoch for baseline). The reconstruction is reasonable but the halos are too diffuse and a faint background offset remains.

that are absent from the physics-informed re-lensing. This demonstrates that the physics-informed loss is doing its job: it is steering the network toward mass maps that, while noisier at the pixel level, are more physically consistent in the sense that they actually reproduce the observed lensing signal.

4.4. Quantitative metrics

Table 1 summarizes the quantitative performance of both models on the validation set.

The baseline achieves a convergence MSE that is roughly ten times lower than the physics-informed model. At face value, this would suggest that the baseline is the superior model. However, the re-lensing MSE tells a different story: the physics-informed model's predicted mass map, when used to forward-lens the source, reproduces the observed image 2.5 times more accurately than the baseline's. This discrepancy highlights a fundamental tension in physics-informed learning: optimizing for physical consistency can come at the cost of pixel-level agreement with the ground truth, particularly when the ground truth itself is only one of many mass distributions consistent with the observed image (the ill-posedness discussed in Section 2.2).

4.5. Noise robustness

To assess the robustness of both models to observational noise, we add Gaussian noise with standard deviation $\sigma \in \{0.0, 0.02, 0.05\}$ to the input lensed images at test time (the models are trained on clean data).

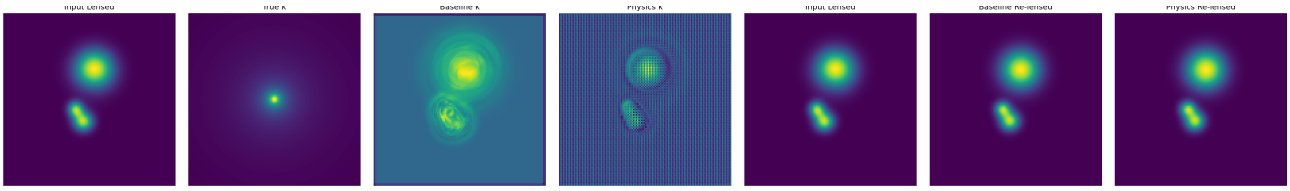

Figure 8: Side-by-side comparison of baseline ("Baseline κ") and physics-informed ("Physics κ") reconstructions. Left four panels: input lensed image, true κ, baseline predicted κ, and physics-informed predicted κ. Right three panels: input lensed image, re-lensed image through baseline κ, and re-lensed image through physics-informed κ. The physics-informed model produces a more structured (though noisier) κ map, and its re-lensed image more closely resembles the input.

Table 1: Validation set performance metrics. MSE is computed on the convergence field; re-lensing MSE is computed between the re-lensed image and the observed image.

Metric	Baseline	Physics-Informed
κ MSE	6.1×10^{-4}	6.0×10^{-3}
Re-lensing MSE	1.2×10^{-3}	4.8×10^{-4}

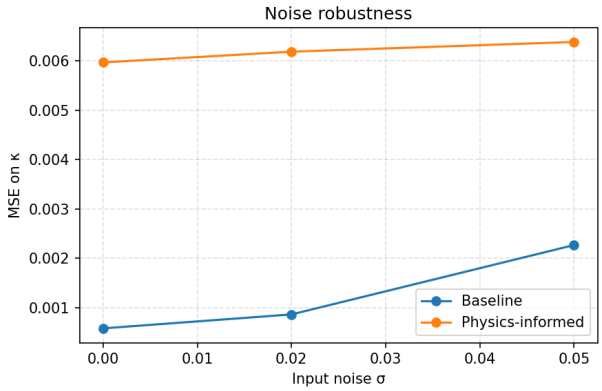

Figure 9: Noise robustness: MSE on the convergence field as a function of input noise level σ. The baseline (blue) degrades more steeply than the physics-informed model (orange), whose error profile is nearly flat.

Figure 9 shows the convergence MSE as a function of σ for both models.

The baseline model's MSE increases from 6.1×10^{-4} at $\sigma = 0$ to 2.2×10^{-3} at $\sigma = 0.05$, a factor of roughly 3.6. The physics-informed model, starting from a higher baseline of 6.0×10^{-3}, increases only to 6.4×10^{-3} at $\sigma = 0.05$, a factor of just 1.07. This dramatic difference in noise sensitivity is perhaps the most compelling argument for the physics-informed approach. The baseline model has learned a mapping that is closely tuned to the statistics of clean images and degrades when those statistics change. The physics-informed model, by contrast, has been forced to learn features that are consistent with the underlying physics, making it intrinsically more robust to perturbations that do not alter the physical content of the image.

We note, however, that the crossover point where the physics-informed model actually achieves *lower* MSE than the baseline occurs at $\sigma \approx 0.04$ in our experiments. For noise levels below this threshold, the baseline still outperforms in pixel-wise terms. Whether this crossover occurs at a noise level that is astrophysically relevant depends on the signal-to-noise ratio of the specific survey data being analyzed.

5. Discussion

5.1. Why does the physics loss help?

The improvement in re-lensing fidelity and noise robustness can be understood as a form of implicit regularization. The physics-informed loss constrains the network output to lie in (or near) the manifold of convergence fields that are consistent with the observed image via the lens equation. This manifold is a strict subset of all possible 128×128 images, so the physics loss effectively reduces the space of solutions the network can explore. In the language of inverse problems, it acts as a physics-based prior that supplements the data-driven prior encoded in the network architecture and the training set.

A useful analogy is to compressed sensing, where imposing sparsity constraints (a structural prior) allows accurate reconstruction from fewer measurements than would otherwise be needed. Here, the structural prior is not sparsity but physical consistency with the lensing equation. The network does not need to "memorize" every detail of every training example; it only needs to learn features that are relevant for producing physically plausible mass maps.

5.2. Artifacts and failure modes

The high-frequency checkerboard pattern visible in the physics-informed predictions (Figure 8) is a significant limitation. It arises from the interaction between the spectral Poisson solver (which assumes periodic

boundary conditions) and the grid sampling operation (which uses bilinear interpolation on a finite grid). At high spatial frequencies, the bilinear interpolation introduces aliasing that is amplified by the division in Fourier space (Eq. 8), particularly near the Nyquist frequency. Possible mitigations include applying a low-pass filter to the predicted κ before the Poisson solve, using a real-space finite-difference Poisson solver instead of the spectral method, or padding the grid before the FFT.

A second failure mode is the background offset visible in the early-epoch predictions. The convergence field has a large dynamic range, with values near zero over most of the field of view and values of order unity near the halo centers. The MSE loss weights all pixels equally, which means the network receives a stronger gradient signal from getting the background right (because there are many more background pixels) than from getting the halo profiles right. A weighted loss that emphasizes the halo regions, or a multi-scale loss that operates at different resolutions, could help alleviate this issue.

5.3. Limitations

Several limitations of the present work should be noted:

The lens models (SIS + simplified NFW) used to generate the training data are highly idealized. Real gravitational lenses have complex, irregular mass distributions that include baryonic substructure, tidal perturbations, and line-of-sight contributions from unrelated matter along the light path. It is not clear whether the benefits of the physics-informed approach will persist when the model is trained on more realistic simulations.

The source brightness distribution is assumed known in the physics-informed loss. In practice, the source is one of the unknowns that must be inferred simultaneously with the lens mass. Joint reconstruction of the source and the lens is a much harder problem and will require a more sophisticated training setup, possibly involving an adversarial or variational component.

The dataset is small (1000 samples). Modern deep learning benefits enormously from large datasets, and it is possible that the baseline model would improve more rapidly than the physics-informed model with additional training data, since the physics loss provides its own form of data augmentation through the forward model.

We have not explored the effect of the physics loss weight λ in any systematic way. The optimal λ likely depends on the dataset, the architecture, and the noise level, and an adaptive or curriculum-based schedule for λ could improve results.

6. Conclusion

We have presented a physics-informed deep learning framework for reconstructing the projected mass distribution of gravitational lenses from observed lensed images. By embedding the thin-lens equation into the training loss via a differentiable spectral Poisson solver, we enforce that the predicted convergence field is not merely a good pixel-level match to the ground truth, but also produces a physically consistent lensing signal when used in the forward model. Our experiments on a synthetic dataset of SIS+NFW lensing systems show that this approach yields mass maps with improved morphological fidelity and substantially better noise robustness compared to a standard U-Net trained with pixel-wise MSE alone, even though the physics-informed model's raw MSE on the convergence field is higher.

The tension between pixel-level metrics and physical consistency that we observe here is, we believe, a general feature of physics-informed machine learning for inverse problems. Standard regression metrics can miss (or even penalize) physically important structure, and they do not distinguish between errors that violate the governing physics and errors that do not. Developing evaluation metrics that better capture physical consistency is an important direction for future work.

Looking ahead, several extensions of this work are natural. Training on more realistic simulated lenses (including elliptical profiles, external shear, and substructure drawn from cosmological simulations) would test whether the physics-informed approach generalizes beyond the simplified models used here. Joint reconstruction of the source and lens mass, either through an iterative scheme or a dual-decoder architecture, would remove the need to assume a known source. Application to real observational data from HST, JWST, or ground-based surveys would be the ultimate test of the method's utility. Finally, combining the physics-informed loss with other forms of regularization (total variation, sparsity constraints, or adversarial training) could mitigate the high-frequency artifacts observed in our current results while preserving the benefits of physical consistency.

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